

ROSTISLAV LYUPA

Backend Developer (Go) ~ C++ Developer
Distributed Systems | High-Load Infrastructure

 @rrrrrusty

 rostislav.a.lyupa@gmail.com

 github.com/lyro41

 Open to relocation:

 /in/rostislav-lyupa

 AM, GE, RS, AE

SUMMARY

Backend/System engineer with 2 years of production experience building high-load FinTech infrastructure in Go, plus a broader 5+ year hands-on background in C/C++ system engineering. Scaled a trading platform through 10x+ volume surges and cut analytics processing time ~20x. Background also includes ICPC Quarter-finalist standing and national olympiad medals.

SKILLS

Languages	Go, C/C++, Python, C#, SQL
Backend & Infra	ClickHouse, PostgreSQL, Protobuf, gRPC, REST API, Docker, Kubernetes
Observability	Prometheus, Grafana, pprof
Tools & GUI	CMake, Git, Linux, Qt, Telegram Bot API

EXPERIENCE

- 08/2023 – 07/2025

Backend Developer (Go)

Profitage

 - Redesigned a single-exchange (Binance-only) integration into a generalized, scalable framework supporting multiple exchanges; built a trading simulation engine and Bybit API integration on top of it using worker pools and proxy rotation, maintaining stability through 10x+ volume surges.
 - Redesigned analytical DB architecture and SQL query patterns, cutting processing time from 30 days to 1–2 days by parallelizing independent subprocesses and eliminating redundant execution; built analytical dashboards to automate reporting and streamline data analysis.
 - Created internal testing tools and API mocks; maintained open-source forks of third-party APIs.
 - Monitored system health and logs, proactively resolving production bugs and connectivity issues.

Go | ClickHouse | Grafana | Prometheus | gRPC | REST API | Protobuf | Kubernetes | Git
- 08/2022 – 09/2022

Gameplay Developer (Freelance)

The Story of Red Cloud

 - Overhauled legacy C# codebase and mentored the team on state machine patterns for complex AI.
 - Resolved critical bugs and physics exploits, unblocking core gameplay for thousands of players.

C# | Git | Code Review | Mentoring

PROJECTS

- C++ | Qt | CMake

QtSIESTA: Scientific Visualization Engine

<https://github.com/lyro41/SiestaChartWidget>

 - Engineered a fully cross-platform coordinator module managing SIESTA simulation pipeline.
 - Built a high-performance real-time parser for streaming molecular modeling data, with a custom 2D chart rendering engine in Qt optimized for high-frequency updates and smooth data plotting.
- C++ | LaTeX

Custom Programming Language & Compiler (Group of 3)

<https://github.com/lyro41/Compiler>

 - Designed and implemented a recursive descent parser and lexical analyzer from scratch.
 - Authored a comprehensive 33-page Language Specification and User Manual, defining and explaining the grammar, control structures, and usage patterns of the Br9h programming language.
- C++ | Qt

Scientific Variable Evaluator

<https://github.com/lyro41/SuperCalculator>

 - Developed a C++ math evaluator with a Qt GUI and a library of math functions.
 - Managed big integers and arbitrary-precision logic by handling mantissa and exponent manually.
 - Programmed the Shunting-yard to parse expressions of variables and numbers into RPN.
 - Developed a recursive system where variables can refer to other variables in their formulas.
- C++ | Qt

Native Desktop Calendar & Task Manager

<https://github.com/lyro41/DesktopCalendar>

 - Built a desktop calendar and task manager using C++ and the Qt framework.
 - Designed a custom UI layout with a feature to change the calendar background.
 - Developed logic for recurring events and local storage for saving tasks.

EDUCATION

2021 – present (on leave)	Bachelor's Degree Applied Mathematics Program	National University of Science and Technology MISIS, Moscow (first two years GPA: 5.00/5.00)
2018 – 2021	High School Diploma Computer Science Track	Kapitsa Phystech-Lyceum (MIPT-affiliated Specialized High School) (GPA: 4.71/5.00)

VOLUNTEERING

2021 – present	Jury Member (Computer Science)	"Start to Innovate" Student Conference, Kapitsa Phystech-Lyceum
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LANGUAGES

English – C2 (EF SET Certified 80/100), Russian – native